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Assignment 04

Question No 01:

Explain firewall and different types of firewalls along with their benefits and limitations?

Ans:

A firewall is security system that monitors and controls network traffic based on predefined rules. It acts as a barrier between trusted internal network and untrusted external networks.

Types of firewalls:

1. **Packet Filtering firewall:** Checks packets of data and blocks based on IP addresses, ports, etc.

- **Benefits:** Fast, simple, efficient for basic operation.
- **Limitation:** Doesn't inspect packet content, vulnerable to spoofing.

2. **Stateful Inspection firewall:** Tracks active connections, makes decisions based on context.

- **Benefits:** More secure than Packet Filtering, understands traffic flow.
- **Limitation:** Can be resource-intensive, complex config.

3- **Proxy Firewall:** Acts as an intermediary, hides internal IPs and network.

- **Benefits:** Good security, content filtering, hide internal IPs.

• Limitation: Can slow down traffic, breaks end-to-end encryption.

4- Next Generation Firewall: Includes intrusion, prevention, application awareness.

• Benefits: Advanced threat detection, app-level control.

• Limitation: Complex, resource-heavy, costly.

Question No 2:

Explain common characteristics of Intrusion Prevention and Detection System along with advantages and disadvantages?

ANS:

IPS/IDS are tools that monitor network/system activity to detect or prevent malicious behavior.

Common Characteristics:

- Real-time monitoring of traffic and logs.
- Signature-based detection known attack pattern.
- Anomaly-based detection unusual behavior.
- Alerting and reporting for suspicious activity.
- Automated prevention (IPS can block traffic immediately).

Advantages:

- Detect attacks in real-time.
- Protect against known and unknown threats.
- Can monitor encrypted traffic with decryption.

Disadvantages:

- Can generate false positives.
- Need constant updates and tuning.
- Might impact network performance.

Question No 3:

Explain VPNs and different types of VPNs?

Ans:

A VPN creates an encrypted tunnel over the internet, allowing secure communication between user and networks.

Types of VPNs:

1. Remote Access VPN: For individuals to access a secure network.

Use case: Employees working from home.

2. Site-to-site VPN: Connects multiple networks securely. Best for ease of use.

Use case: Branch offices connecting to HQ.

3- Mobile VPN: For mobile devices on the go -
Maintains secure connections while switching networks -

Use case: Secure browsing on public Wi-Fi.

4- Cloud VPN: Connects users to cloud resources

Securely.

Use case: Best for Businesses using cloud services -
